



The step forward Onboard DC Grid

The new way with Onboard DC Grid

Imagine a ship with an efficient and modern propulsion system. It is electric. It has state-of-the-art propellers and thrusters chosen from a variety of available designs (electrical/mechanical), it has the most advanced converters for smooth and efficient speed control, it has modern engines with common rail injection and it can be fuelled by gas. **Take this ship and reduce the specific fuel consumption and emissions by up to 27%, that is today's ship with Onboard DC Grid.**

ABB, the frontrunner in electric propulsion, takes a significant step forward by launching the Onboard DC Grid. Developments in power distribution technology have made this new concept fully practical.

The Onboard DC Grid is a step forward in optimized propulsion and is an extension of multiple DC links that already exist in propulsion and thruster drives. Onboard DC Grid enables us to combine the advantage of AC components with a new smart DC distribution. Just as variable speed drive allows the electric propulsion motors to be run at their optimum working point, Onboard DC Grid allows the diesel engines to run at variable speed for top fuel efficiency at each load level. And, the Onboard DC Grid enables full flexibility in combining energy sources, including renewables.

To the ship owner Onboard DC Grid means:

- Up to 27% reduction of specific fuel consumption
- Less maintenance of generator sets
- Improved dynamic response and maneuverability
- Increased space for payload
- Ready for new energy sources

To yard and designer Onboard DC Grid means:

- Fewer components to be installed
- Reduced equipment footprint and weight
- Easier cable installation
- More flexible placement of components

Design principles

Factors that have led ABB's designers to adopt the alternative method of DC power distribution are:

- By far the largest proportion of the total power load on board a vessel is for propulsion and thrusters. This power must be processed as a DC input to the variable frequency inverter that performs the speed control of the motor. Actually distributing at a DC level allows the losses in a switchboard and transformer to be eliminated from this power flow.

- When diesel-electric generators run at constant r.p.m. fuel efficiency is compromised. By distributing electricity in DC it is no longer required to maintain a fixed electrical frequency. Consequently, the engines can run at variable speed and therefore at the best possible operating point to achieve the best fuel efficiency along the full power range.

Optimization and ruggedness

Each power source and consumer on the Onboard DC Grid is an AC or DC "island" and the only connection between them is the DC bus. This yields two advantages:

- Each power source and consumer can be controlled and optimized independently.
- Complex interactions that can arise between units that share an AC connection will never occur. Consumers fed by the Onboard DC grid are designed not to interact even under fault conditions.

Energy storage

In the Onboard DC Grid solution, energy storage may be included to improve the system's dynamic performance. Diesel engines are slow to handle large, quick load changes. By using batteries or super capacitors to provide power for a short time, the ship's control capabilities can be improved. This will benefit especially vessels with Dynamic Positioning. Energy storage can also be used to absorb rapid power fluctuations seen by the diesel engines, thereby improving their fuel efficiency.

Fit for the future

The Onboard DC Grid is an open power platform which easily allows reconfiguration in number and types of power sources and consumers, power levels and other modernizations. Alternative energy sources that will become significant during a typical vessel lifetime of 20+ years will be easier to adopt in a vessel with Onboard DC Grid because they will not be bound to a AC system, nor will they require redesign of a main switchboard. To the ship owner this will mean a more flexible and competitive vessel.

Dina Star

ABB's first delivery of the DC grid was to Dina Star – a 93 meter long and 5000 DWT multipurpose offshore supply and construction vessel operated by Myklebusthaug Offshore.

About Dina Star

Platform Support Vessel (PSV) Dina Star, delivered to Norwegian offshore owner Myklebusthaug Offshore by Kleven Yard in April 2013, is the first powered with ABB's full Onboard DC grid system to provide fuel efficiency and lower emissions. Dina Star with its Onboard DC Grid technology was nominated for the 2013 Energy Efficiency award at Nor-shipping.

Vessel name

Dina Star

Vessel type

Platform Supply Vessel (PSV)

Owner

Myklebusthaug Offshore, Norway

Yard

Kleven Shipyard, Norway

Class

DNV

Year of delivery

2013

ABB's solution and scope

- Onboard DC Grid system, including all power, propulsion, automation and advisory systems
- EMMA™
- Onboard DC Grid
- Automation solution PEMS



The solution



In its simple way the Onboard DC Grid is just an extension of the already multiple DC-links already existing in all propulsion and thruster drives accomplishing for usually more than 80% of the electrical power consumption on electric propulsion vessels.

This extension means that we keep all the good and well proven products already used in today's electric ships like AC generators, inverter modules, AC motors, etc. All main AC SWBDS and transformers are however no longer needed and you have the most flexible power and propulsion system to date. The main innovations with this new Onboard DC Grid are the design and control of the protection system and optimized energy flow.

This technical note describes the design and configuration of the Onboard DC Grid system, with a discussion of the various benefits.

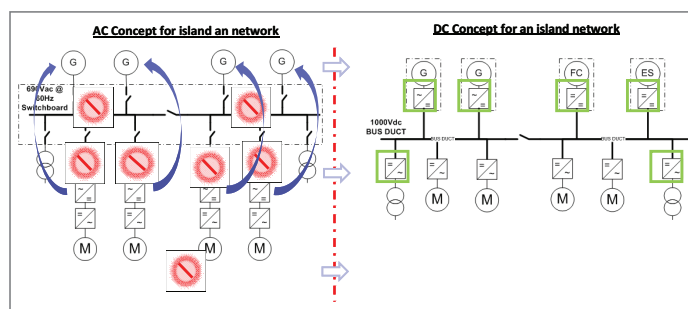


Fig. 1 From AC to DC (single line old vs. new design)

Onboard DC Grid

There are several ways of configuring the Onboard DC Grid from a multidrive approach (figure 3) to a fully distributed system (fig. 4 next page). In the multidrive approach all converter modules are located in the same lineup within the same space layout as today's main AC switchboard. For the distributed system each converter component is located as near as possible to the respective power source or load.

Common for both alternatives is that the main AC SWBD and all thruster transformers are omitted in the new concept. Instead all generated electric power is fed directly or via a rectifier into a common DC bus that distributes the electrical energy to the consumers. Each main consumer is then fed by a separate inverter unit. The 220V AC distribution (e.g. "hotel load") will be fed using island converters, specially developed to feed clean power to these more sensitive circuits. Further converters for energy storage can be added to the grid. This energy storage could for example be batteries or super capacitors for leveling out power variations.

The main benefits of this approach is a reduction of the fuel consumption - up to 27% reduction of spesific fuel oil consumption. In addition the system allows for considerably weight and space savings thus leading to increased cargo capacity.

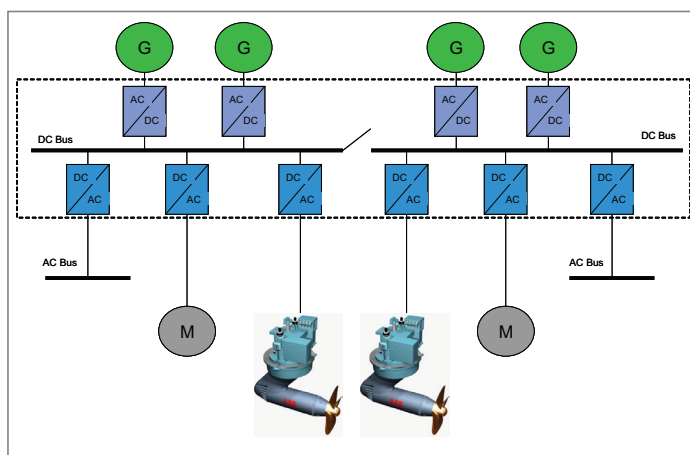


Fig. 2 Onboard DC Grid; Multidrive approach.

The major breakthrough in the development of the DC grid was the development of an innovative protective system. AC currents are by nature far simpler to break because of their natural zero crossing every half cycle. DC circuit breakers do exist to some extent but are more complex, larger and more expensive than comparable AC circuit breakers.

By designing the Onboard DC Grid we have looked at the whole concept and layout from a totally new perspective, following two main principles:

- Equipment shall always be protected.
- Proper selectivity shall be ensured in such a way that safe operation is maintained after any single failures.

Onboard DC Grid is a new electric power distribution concept that, while utilizing the well proven AC generators and motors, opens new opportunities for efficiency improvements and space savings. The efficiency improvement is mainly accomplished by the fact that the system is no longer locked at a specific frequency (usually 60Hz on ships), even though any 60Hz power source also would be connectable to the Grid. The new freedom of controlling each power consumer totally independently opens up numerous ways of optimizing the fuel consumption. Today almost all energy producers on electric ships are combustion engines, most operating on liquid oil (HFO/MDO), some on gas (from LNG mainly), and even some with Dual Fuel capability (liquid fuel or gas).

When operating these engines at constant speed the fuel consumption is lowest at a very small operating window around 85% of rated load. By operating at variable speed it is possible to select the optimal working point within the full engine envelope (figure 5. next page)

In the most distributed version of the Onboard DC Grid, each power converter is located as close as possible to the respective consumer or producer. Each production unit has the possibility of an integrated rectifier mounted directly on the unit itself or alternatively in a separate cabinet close by. There are no needs of collecting all these units in a centralized “switchboard room” as in a classic design.

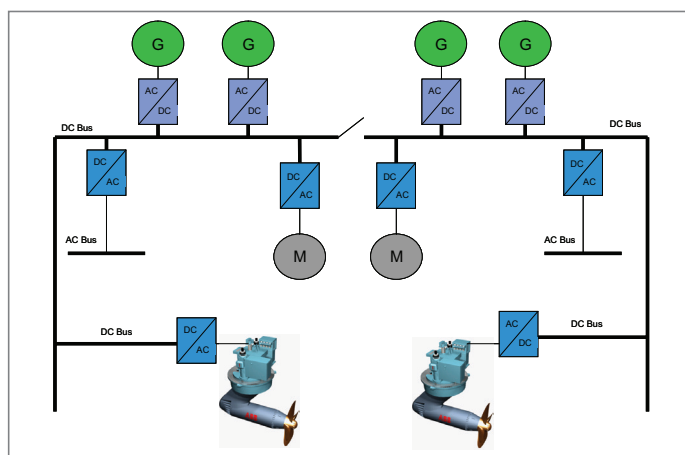


Fig. 3 Onboard DC Grid; Distributed approach.

Since the main AC SWBD with its AC circuit breakers and protection relays is omitted from the new design, it has been essential to devise a new protection philosophy that fulfills class requirements for selectivity and equipment protection. In doing so it has also been a key requirement to minimize use of expensive and space consuming DC circuit breakers.

Proper protection of the Onboard DC Grid is achieved by a combination of fuses and controlled turn-off of semiconductor power devices. Since all energy producing components have controllable switching devices (either thyristor rectifier for AC producers and DC/DC converters for DC producers) the fault current can be blocked much faster than what is possible with traditional circuit breakers with associated protection relays.

Exploring the benefits

Onboard DC Grid is here now; it combines the best of both AC and DC components/systems available, it is fully compliant with rules and regulations, and is the choice for the future with low emission and low fuel consuming ships.

Exploring the benefits

Efficiency

Figure 5 shows the spesific fuel oil consumption map of a diesel engine. It can be clearly seen from this picture that it is possible to run the engine with the lowest possible fuel consumption at least down to 50% loading. This is especially beneficial for vessels operating in Dynamic Positioning, where average electric thruster loads are normally low due to low propeller speeds and normal weather conditions, but number of running engines is higher than really needed because of safety reasons. The pure electrical efficiency will also contribute to the improved efficiency with less installed components (no main switchboard and thruster transformers).

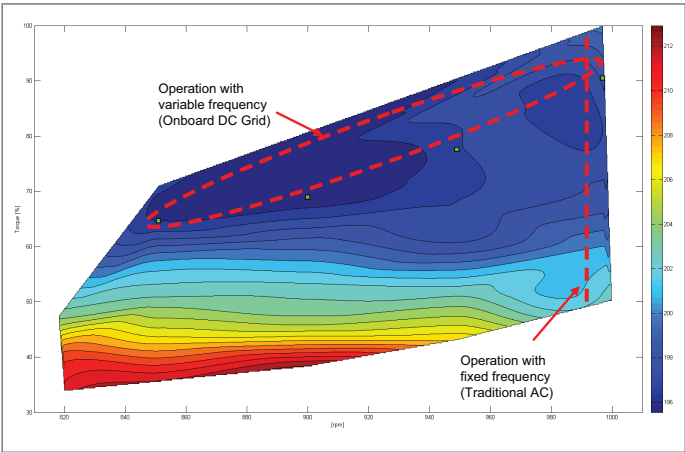


Fig. 4 Engine fuel tests at variable speed (color scheme indicates Specific Fuel Oil Consumption (SFOC) in g/kWh .

However, the biggest fuel savings potential lies in the ease with which energy storage devices, like batteries or super capacitors, can be added to the system. In this area the technology has developed quite much in the last decade and is expected to develop further. Energy storage will help the engines level out load variations from the thrusters and other large loads, following the same principle used by hybrid cars. Depending on the operational profile, energy storage can further improve fuel efficiency as much as 27%.

Weight and space arrangement

Another benefit with Onboard DC Grid is the reduced weight and footprint of the installed electrical equipment. Table 1 outlines a comparison between DC Grid and a traditional AC system for a Plaform Supply Vessel (PSV).

Operations

Onboard DC Grid enables new ways of thinking for operational optimization. As the system is flexible by combining different energy sources like engines, turbines, fuel cells, etc., there is a huge potential for implementing a real energy management system, taken into account varying fuel prices and availability of different fuel. This kind of optimization may be some years ahead, but with Onboard DC Grid the vessel is prepared for the future.

Equipment	Q'ty	Rating	Weight Traditional	Weight Onboard DC Grid
Generators w/aux	4	2500 kVA	38000 kg	39000 kg
Main AC SWBD	1	690 VAC	4450 kg	0 kg
Main DC distribution	1	1000 VDC	0 kg	2400 kg
Distribution AC	1	450V/230V	14490 kg	16530 kg
Propulsion drives	2	3500 kVA	31980 kg	13680 kg
Thruster drives	3	1200 kVA	26600 kg	13750 kg
Total			115520 kg	85360 kg

Table 5. Comparison of weights for installed electrical equipment for an example: PSV. Traditional AC concept vs. Onboard DC Grid.

What is available today and would help in solving the traditional challenge for DP operation is the fuel efficient running of engines at part load. In the most severe DP operations today the electrical plant is operated by a minimum of 2-split

configuration for safety reasons. This gives the vessel possibility to keep its position even if one side of the power plant is failing. However, running in split mode does not utilize the full benefits of electric propulsion in general as a total optimization of running engines is not possible. With Onboard DC Grid the split mode operation can be run more efficiently as the engine speed can be adjusted and optimized to the required load without the need for changing the number of generators online.

Protection and safety

As already mentioned, the protection philosophy is based on a combination of fuses and controlled switches. In short; fuses are used to protect and isolate inverter modules in case of serious module faults. This is no different to current LV frequency converters. In addition, input circuits separate the inverter modules from the main DC bus and afford full control of reverse power, both in fault and normal conditions (as for example in propeller braking mode). This means that faults on a single consumer will not affect other consumers on the main DC distribution system. In the event of severe faults on the distributed DC bus, the system is protected from generators by means of a controllable thyristor rectifier which also doubles as a protection device for the generator. Isolators are installed in each circuit branch in order to automatically isolate faulty sections from the healthy system.

In sum, the Onboard DC Grid fully complies with rules and regulations for selectivity and equipment protection. Further; any fault current will be cleared within maximum 40ms. This results in a drastic reduction in Onboard DC Grid fault energy levels as compared with traditional AC protection circuits where fault durations can reach up to 1s.

Concluding remark

The Onboard DC Grid system is a new way of distributing electricity on board ships. It can be used for any electrical ship application up to at least 20MW and operates at a nominal voltage of 1000V DC. The power distribution can be arranged with all cabinets in a single line up (multidrive approach) or distributed throughout the vessel by short-circuit proof DC busbars.

Benefits for the shipowners and operators

- Up to 27% specific fuel saving if taking full advantage of all features including energy storage and variable speed engines.
- Reduced maintenance of engines.
- Improved dynamic response by use of energy storage, which may give a better DP performance with lower fuel consumption or more accurate positioning.
- Increased space for payload through lower footprint of electrical plant and more flexible placement of electrical components.
- More functional vessel layout through more flexible placement of electrical components.
- A system platform that affords simple “plug and play” retrofitting possibilities to adapt to future energy sources
- Considerable noise reduction (up to 30%).

Benefits for shipyards and designers

- More flexible placing of electric components.
- Reduced footprint and weight of electrical equipment by up to 30%.
- Less cabling and cabling connection, by means of reduced number of components and use of bus ducts.

Onboard DC Grid is here now; it combines the best of both AC and DC components/systems available, it is fully compliant with rules and regulations, and is the choice for the future with low emission and low fuel consuming ships.

Onboard DC Grid got approval in principle by ABS

Approval in principle confirms Onboard DC Grid's suitability for marine application. The new system supports the growth of electrical propulsion systems onboard ships.

Classification Societies are organizations within the maritime industry that establish and maintain technical standards for the construction and operation of ships and offshore structures, whose main focus is the safety of life at sea and protection of the environment.

AIP is a framework used by most classification societies to review and approve innovative and novel concepts not covered by traditional classification rules. In order to be granted the AIP, the Onboard DC Grid was subject to a series of risk assessment techniques to determine if the concept provides acceptable levels of safety in line with current marine industry practice, requirements and standards.

The AIP granted by ABS is a strong tool to support and streamline the design, approval, execution and inspection of new projects. Therefore, the benefits are extended to all parties involved on the shipbuilding process: designers, ship-owners, ship-yards and classification society.

Contact us

For more information, please contact:

ABB AS

Stålhaugen 5
6065 Ulsteinvik
Norway

www.abb.com/marine

LM062014 © Copyright 2014 ABB. All rights reserved.